

Leveraging Data Analytics and Behavioral Analysis for Improved Testimony and Incident Correlation

Incident Correlation and Pattern Identification

Project ID: R25-020

Project Proposal Report

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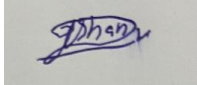
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Declaration

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of Higher Learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

Criminal investigations often involve analyzing vast amounts of data to identify trends, connections, and key insights. Traditional investigative methods rely heavily on human effort, making them time-consuming and prone to errors. This work presents an advanced approach utilizing machine learning algorithms to enhance incident analysis and case correlation by examining multiple dimensions, including crime types, locations, and individuals involved.

One of the core features of the proposed system is semantic analysis, which plays a crucial role in investigating discrepancies and similarities in witness statements. By leveraging natural language processing (NLP), the system can detect inconsistencies, helping investigators assess credibility and identify patterns in testimonies. This feature significantly enhances the accuracy of investigations by reducing reliance on subjective human judgment.

Additionally, the system incorporates an incident alert mechanism, which proactively identifies anomalies and significant trends. By detecting outliers and emerging patterns, the system helps law enforcement agencies prioritize critical leads and allocate resources efficiently. This real-time alerting capability enhances response time, ensuring that potential threats and high-priority cases receive immediate attention.

The effectiveness of this machine learning-driven approach has been evaluated through extensive testing, comparing it with conventional investigative techniques. The results indicate notable improvements in accuracy, efficiency, and overall case resolution times. By integrating this technology into existing investigative workflows, law enforcement agencies can minimize manpower requirements, optimize resource allocation, and significantly enhance crime analysis. Implementation challenges, such as data security, system integration, and legal considerations, have also been examined to provide practical guidelines for seamless adoption. This work represents a major step toward modernizing crime investigations, making them more data-driven and less susceptible to human error.

Keywords: Machine Learning, Crime Investigation, Incident Analysis, Semantic Analysis, Case Correlation, Incident Alert System, Data-Driven Law Enforcement

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INTRODUCTION

1.1 Background Study

Criminal investigations rely increasingly on case analysis, a tedious, inefficient, and error-ridden activity that involves real-time cross-matching between cases and witness statements, with a high probability of missed opportunity and extended case resolution times. To counteract such a problem, our AI-based solution utilizes incident and pattern algorithms in machines, thus optimizing investigation efficiency.

The Incident Correlation and Pattern Identification module identifies new incidents in relation to case background specifically, and cases, commonalties, and discrepancies that traditional approaches miss, in a focused manner. By leveraging sophisticated processing algorithms, the system helps investigators detect case relationships, detect potential offenders, and make inferences about events in terms of location, participants, and crime.

Some of its key capabilities include clustering and classification algorithms for smart incident-to-past case correlation, semantic analysis for witness testimony comparison, and an alert function for notifying investigators about important relationships. Real-time information insights are incorporated in the module, minimizing traditional analysis for patterns and supporting rapid fact-based decision-making with increased accuracy for law-enforcement agencies.

Unlike conventional investigative tools that focus on isolated data points, this component provides a holistic approach by dynamically analyzing diverse datasets, ensuring a comprehensive perspective on case patterns. By streamlining criminal investigations and reducing human error, the Incident Correlation and Pattern Identification component plays a vital role in modernizing law enforcement processes and improving case resolution rates.

1.2 Literature Survey

Incident correlation and pattern discovery have become critical aspects in modern policing, with traditional investigation techniques often hindered by information complexity and bulk in criminal cases. Emerging technology in artificial intelligence (AI) and machine learning in recent times has unlocked new avenues for incident relation-finding operations automation and improvement, lessening the use of cross-matching in a manual manner and enhancing overall investigation efficiency. AI use in investigations has received significant traction, with an objective of computerizing information analysis to detect trends and relations in cases. AI-facilitated forensic tools have been designed to scan long lists of case information, including witness statements, suspect profiles, and crime site reports, in a quest to detect recurring trends in criminal cases, according to Johnson et al. (2023) [1]. The study showed that such tools could effectively streamline investigation times through cross-matching with similar cases, and in the process, expose previously concealed relations between incidents.

Machine learning algorithms such as clustering and classification have been researched extensively with an aim of discovering patterns in crime analysis. In a study by Lee et al. (2022) [2], neural networks were utilized to reveal relationships between crimes and geographical locations, offender profiles, and types of crime. In the study, it showed that machine algorithms increased future prediction accuracy of suspects and connecting cases that seemed disparate in nature. This study validates that AI-powered algorithms for correlation can boost the ability of law enforcement to detect trends in crime and future occurrences. Natural language processing (NLP) technology has become a critical tool for unstructured information analysis, including statements and testimony, and police reports. In a study performed by Smith et al. (2021) [3], NLP application in information extraction in investigation files was a key issue, and comparative analysis of statements between cases facilitated through it. By employing such an approach, biases and inaccuracies in information collection through humans are reduced, and uniform and fact-based information about relationships between cases is guaranteed.

The integration of artificial intelligence with real-time processing of information has proven effective in law enforcement. Brown et al. (2024) [4] conducted a review of developing a real-time crime monitor utilizing AI analysis of real-time incident reports, CCTV footage, and emergency dispatch transcripts. In their work, they emphasized that such a system enables investigators to react in a proactive manner towards impending danger through immediate identification of repeat patterns of crime. Progress in personalized AI tools has increased efficiency in analyzing crime. Zhang et al. (2023) [5] analyzed the contribution of adaptive AI algorithms, which tailor their correlation algorithms in relation to individual requirements in law enforcement. These algorithms have the capability to learn new cases in real-time, subsequently enhancing their capabilities in identifying trends and predicting case links over a period of time. According to the study, such adaptable systems contribute a lot towards enhancing investigation output through reduced workloads and increased case closing rates. AI, machine learning, and natural language processing integration in criminal investigations have revolutionized incident correlation and trending analysis. According to studies, efficiency, accuracy, and uniformity in

identifying trends in crime can be increased through use of automated systems, and such a move reduces reliance on conventional, tedious techniques. As technology continues, AI-powered solutions will have an increasingly significant role in transforming law enforcement, subsequently paving the way for effective approaches in solving crime.

1.3 Research Gap

The integration of artificial intelligence (AI) in criminal investigations has significantly advanced, focusing on areas such as automated evidence analysis, incident correlation, and AI-driven pattern recognition. However, current systems still face several limitations in effectively linking and visualizing crime patterns, leading to inefficiencies in investigative processes. While various studies have explored AI applications in crime detection and forensic science, there are critical gaps in leveraging real-time data processing, automated correlation, and graph-based visualization for enhanced crime analysis.

One of the key challenges in existing research is the lack of real-time case data analysis. Most AI-driven crime investigation tools operate on historical data, making it difficult to detect and respond to evolving crime patterns instantaneously. Studies emphasize AI's role in forensic analysis and deception detection [1], [2], yet few have explored its application in real-time monitoring and predictive analytics for crime incidents.

Furthermore, automated incident correlation remains limited in current research. Existing approaches rely on manually linking similar cases based on predefined criteria, which can overlook hidden relationships between incidents. Some studies have investigated AI techniques for identifying crime patterns [3], [4], but they lack robust automation mechanisms for correlating incidents dynamically across different datasets.

Another significant gap is the absence of AI-powered pattern recognition models specifically designed for criminal investigations. While AI has been widely used in fraud detection and cybercrime analytics [5], [6], its application in recognizing complex crime patterns involving multiple suspects, locations, and timelines remains underdeveloped. Current research mainly focuses on isolated cases rather than developing comprehensive AI models capable of detecting interrelated criminal activities.

Additionally, graph-based visualization of incident links is rarely addressed in forensic AI research. Traditional crime mapping tools primarily use geographical representations without interactive link analysis. While network analysis has been applied in social sciences [7], [8], its integration with AI-driven crime analytics for visually representing relationships between cases, suspects, and evidence is still limited.

Moreover, existing crime investigation tools lack a fully integrated web platform that combines multiple AI-driven capabilities into a single, accessible interface. Research efforts have mostly been fragmented, focusing separately on deception detection, predictive modeling, and forensic AI [9], [10]. However, there is little work on unifying these elements into a centralized system

that streamlines investigative workflows and enhances accessibility for law enforcement agencies.

Features	Research Paper [1]	Research Paper [2]	Research Paper [3]	Research Paper [4]	Proposed System
Real-Time Case Data Analysis	✗	✗	✓	✗	✓
Automated Incident Correlation	✗	✓	✓	✗	✓
Pattern Recognition using AI	✗	✓	✓	✗	✓
Graph-Based Visualization of Incident Links	✗	✗	✗	✓	✓
Integrated web platform	✗	✗	✗	✗	✓

1.4 Research Problem

A lot of data is present in the realm of crime investigations which comprises police reports, witness comments, forensic evidence, and historical case records. They traditionally have depended on manual techniques to look up incidents, find patterns, and connect crimes to each other. But such an approach eats up time, is prone to human error, and leads to overlooked details that could be critical in solving cases.

One of the major difficulties in modern law enforcement is the effective correlation of incidents and the extraction of concealed patterns that connect crimes over various periods and regions. Existing investigation tools either concentrate on specific tasks (biometric recognition, crime mapping, or forensic analysis) or are not a consolidated tool that automatically analyses case data so that links between incidents can be revealed. Law enforcement agencies face the challenge of handling vast, unstructured datasets and gaining actionable insights from them without an AI-powered solution, which prolongs case resolutions and enables criminals to escape the law.

In addition, unstructured key data sources such as testimony, case notes, and interview transcripts go underused due to the challenge of manually extracting and comparing some key information. However, existing crime analysis systems fall short of using advanced NLP models that can detect context and differences in testimonies. Beyond obstacles with crime data visualization, investigators struggle to keep track of relationships between cases.

Research Question

- How could something like machine learning enhance the identification of patterns between new events and existing data?
- Can semantic analysis identify discrepancies in witness testimonies?
- How can an alert system prioritize critical patterns for investigators?

2. OBJECTIVES

2.1 Main Objective

Incident Correlation and Pattern Identification System is a system which is designed to create advanced machine learning based solution for criminal investigation which remove duplication of countermeasures to interpolation between new events and historical case information. Conventional investigative processes often involve tedious manual cross-referencing that is time-consuming and subject to human error, making it ineffective in uncovering hidden relationships between cases. The system will be able to analyze massive datasets in real time to help law enforcement identify connections between crimes, expose potential suspects and recognize emerging trends with better accuracy.

This saves investigative time by automating case linking and identification of patterns across various different criminal activities, in turn placing less dependence on human effort, process flows and improving the reliability of solving crimes. Additionally, the system's real-time alerts and data-backed insights ensure that investigators are notified in a timely manner about potential case linkages, thus accelerating case closures. Bringing the Future of Crime Fighting Digital age consultancy Digital Crime Detection Digital Evidence Analysis, helps expedite the process by using a data-driven approach for analyzing evidence to solve criminal activity and minimize the chances of reoccurrence.

2.2 Specific Sub Objectives

- **Comparison and Analysis of Patterns** The purpose of this study is to build sophisticated algorithms for machine learning that can analyze incident data, detect recurring trends, and contrast new cases with past experiences.

By employing statistical techniques and AI-facilitated models, the system will detect similar types of offences, suspicious activity, and geographical trends. By having such a mechanism in place, investigators will be aided in discovering concealed relationships between cases, and in consequence, a data-intensive investigation will become a norm for investigations.

- **Classification and Clustering Models**

The system will utilize clustering and classification algorithms to group incidents according to their descriptors, such as type of offence, location, and timing of incident. Unsupervised algorithms, such as K-Means clustering, will group similar cases, and supervised algorithms, such as Random Forest and Support Vector Machines (SVM), will classify new cases according to predefined trends. With such a mechanism, law enforcement will detect hotspots for offences, differentiate between trends in offenders, and effectively reallocate manpower for efficient policing.

- Incident Alert System

The objective of this project is to create an incident alert system with the capability to inform investigators about suspected trends and relations with regard to criminal activity in real-time during events unfolding. By ongoing analysis of information regarding received cases, new cases with similar trends to previous crimes can be predicted and detected, with relevant information shared with investigators in a timely manner. Alerts will enable proactive intervention in trends in crime, contribute towards minimizing repeat victimization, and enable effective resolution of cases.

3. METHODOLOGY

3.1 Requirement Gathering and Analysis

The process of information collection and analysis forms a critical part of creating the Incident Correlation and Pattern Identification System, with a view towards customizing the system according to specific requirements for forensic analysis and law-enforcement agencies. By combining collection and analysis of information from web databases and localized sources, the system aims at providing a thorough picture of trends in crime, investigation-related concerns, and system requirements.

To present a complete picture, information will be collected through the following sources:

1. **Online Databases** – Government databases for criminal activity, academic studies, and articles regarding artificial intelligence in relation to investigations will yield publicly accessible datasets, reports, and case studies regarding investigations and crime. Historical datasets with information regarding trends, patterns, and criminal activity will become accessible through these sources and will form a part of training artificial intelligence and algorithms.

2. **Local Organizations** – Having an ongoing liaison with investigating, forensic, and law-enforcement agencies will allow access to real-life case information and expert analysis. Interviews and questionnaires will be conducted with police officers, detectives, and professionals working in the field of crime analysis in order to clarify case correlations, assess witness testimony, and understand trends in criminal activity.

3.2 Testing and Validation

Following the development of the Incident Correlation and Pattern Identification System, an exhaustive testing and validation exercise will be performed to evaluate its accuracy, dependability, and effectiveness in real-life investigation scenarios. Included in this exercise will be:

- **Unit Testing:** Individual system modules, including the pattern model, testimony analysis through natural language processing, clustering algorithms, and incident alert processes, will receive thorough testing to confirm correct function and delivery of desired output.
- **Integration Testing:** Interaction between system modules will be evaluated to confirm unproblematic information flow, correct case correlation, and timely delivery of warnings. This will enable assurance of integration and effective case information analysis derived from a variety of sources.
- **User Acceptance Testing (UAT):** Law enforcement, forensics, and investigation professionals will evaluate the system through mock case studies. User feedback will be collected in an evaluation of usability, effectiveness in discerning crime patterns, and overall value in enhancing investigations in preparation for full-fledged implementation.
- **Validation:** Correctness of incident correlation and pattern identification performed by the system will be evaluated through comparison with case resolution and analysis performed in the past through expert investigators.

3.3 Deployment and Maintenance

Once testing is successfully completed, the system will be deployed and made available for use by law enforcement agencies. The deployment phase involves installing the proposed system on the existing investigative platforms and setting up a dedicated support desk to assist users. Regular maintenance, including annual and special updates, will be conducted to address issues, integrate user feedback, enhance system performance, and ensure continued accuracy and efficiency in crime analysis.

- **Training:** Law enforcement personnel and investigative teams will be trained on how to effectively utilize the system for incident analysis, case correlation, and real-time alert monitoring.
- **Support and Maintenance:** Continuous maintenance will be provided to ensure the system remains up to date with user requirements and evolving technological advancements, improving reliability and operational efficiency.

3.4 Tools, Technologies and Algorithms

- **Frontend: React** – Used for designing and developing an intuitive and user-friendly interface, enabling investigators to interact seamlessly with crime data visualizations, case correlations, and alerts.
- **Backend: Python** – Handles the core application logic, including pattern recognition, NLP-based testimony analysis, and machine learning models for crime correlation.
- **Database: PostgreSQL / MongoDB** – Stores structured and unstructured case data, including police reports, witness testimonies, crime metadata, and machine learning outputs.
- **Web Frameworks: Django & Flask** – Facilitates API development for integrating backend functionalities with the frontend interface and external law enforcement systems.
- **Cloud Services:**
 - **Amazon Web Services (AWS)** – Provides cloud-based storage, hosting, and computational scalability for large-scale crime data analysis.
 - **Google Cloud AI** – Supports machine learning model training and deployment for predictive crime analysis.
 -
- **IDE (Integrated Development Environment): VS Code** – Used for coding, debugging, and version control.
- **Version Control: GitHub** – Enables collaborative development, version management, and secure code deployment.
- **Machine Learning and NLP Algorithms:**
 - **BERT (Bidirectional Encoder Representations from Transformers)**: Used for semantic analysis of witness testimonies to detect contradictions and correlations across multiple cases.
 - **K-Means Clustering**: Helps in grouping similar criminal incidents based on patterns in crime type, suspect profiles, and geographic locations.
 - **Random Forest & SVM (Support Vector Machine)**: Used for classification and anomaly detection in crime data.
 - **Graph Neural Networks (GNNs)**: Applied for crime network analysis, helping in identifying links between multiple cases.

TECHNOLOGIES TO BE USED



Figure 1.Tools & Technologies

3.5 System Overview Diagram

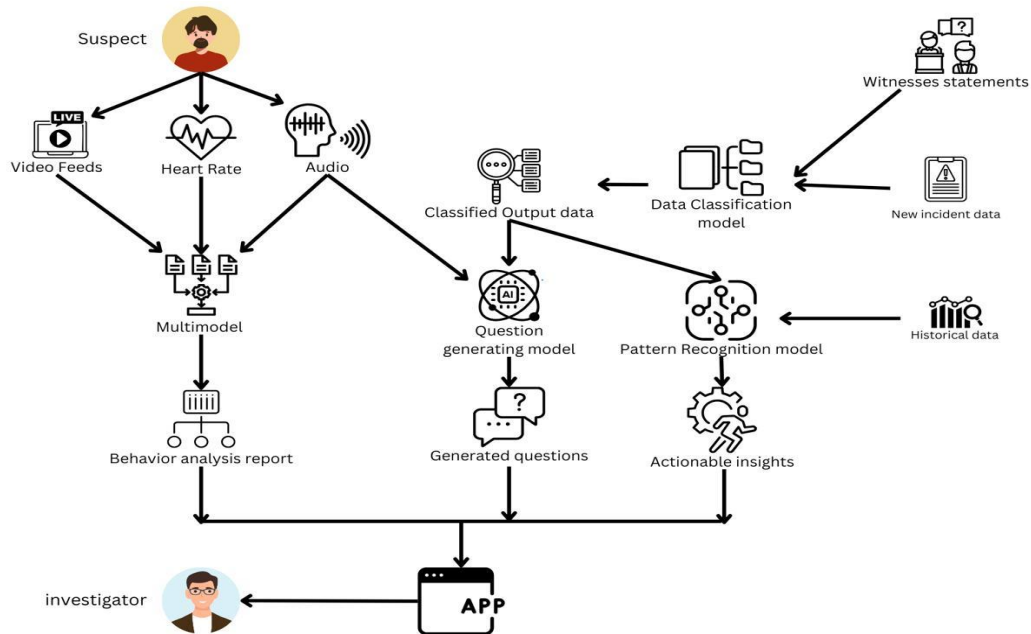


Figure 2. System Diagram

The proposed AI-driven solution enhances criminal investigations through a multi-modal approach that integrates advanced technologies for data collection, testimony evaluation, and incident correlation. The system processes live video feeds, heart rate monitoring, and audio data from suspects during interviews to analyze behavioral cues and detect signs of stress or deception.

Key components of the system include:

- 1. Automated Data Collection and Management:** This component uses OCR and NLP to automatically extract and organize relevant information from police reports, witness statements, and other documents, reducing human error in data handling.
- 2. Behavioral and Emotional Analysis:** The system analyzes video feeds, heart rate monitoring, and audio data to detect emotional and physiological cues, such as stress or discomfort, providing real-time feedback on the credibility of testimonies.
- 3. Dynamic Question Generation and Response Analysis:** NLP techniques are used to analyze responses during interrogations, generating contextually relevant follow-up questions in real-time, enhancing the quality and depth of interviews.
- 4. Incident Correlation and Pattern Identification:** Machine learning algorithms compare new incidents with historical case data to identify patterns and similarities, helping investigators uncover critical connections and insights.

These outputs are delivered to investigators through an intuitive app, streamlining the investigation process and reducing human error. The integrated system offers comprehensive, real-time support, enabling more efficient, accurate, and timely criminal investigations by automating key tasks and providing actionable insights.

3.6 component Overview Diagram

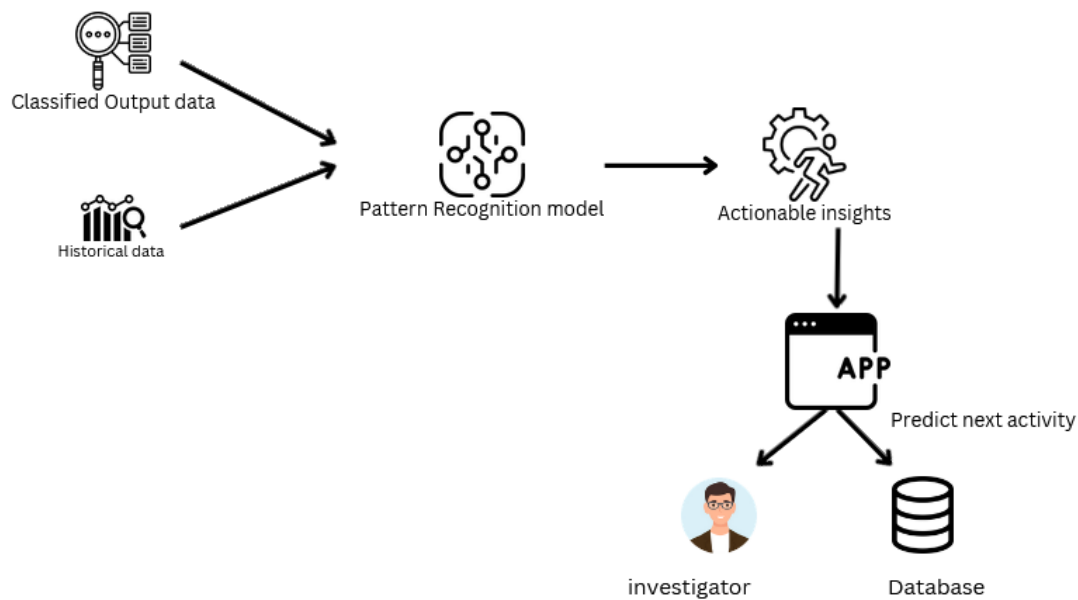


Figure 3. Component Diagram

The Incident Correlation and Pattern Identification system begins with data collection, where historical case records and classified output data are gathered to analyze incidents. The system then proceeds to the Pattern Recognition phase, where machine learning algorithms compare new incidents with past cases to detect similarities, correlations, and emerging trends. By recognizing recurring patterns, the system helps investigators uncover hidden connections between cases. Once patterns are identified, the Actionable Insights phase provides investigators with key findings that assist in decision-making. These insights are integrated into an application that allows investigators to predict the next possible activity based on historical trends.

The system also updates a database to continuously learn from new cases, enhancing its predictive accuracy over time. Finally, the insights generated empower investigators to identify potential threats, detect fraud, and prevent future incidents, ensuring a more proactive and data-driven approach to incident management.

3.7 Gantt Chart



Figure 4. Gantt Chart

3.8 Work Break Down Structure

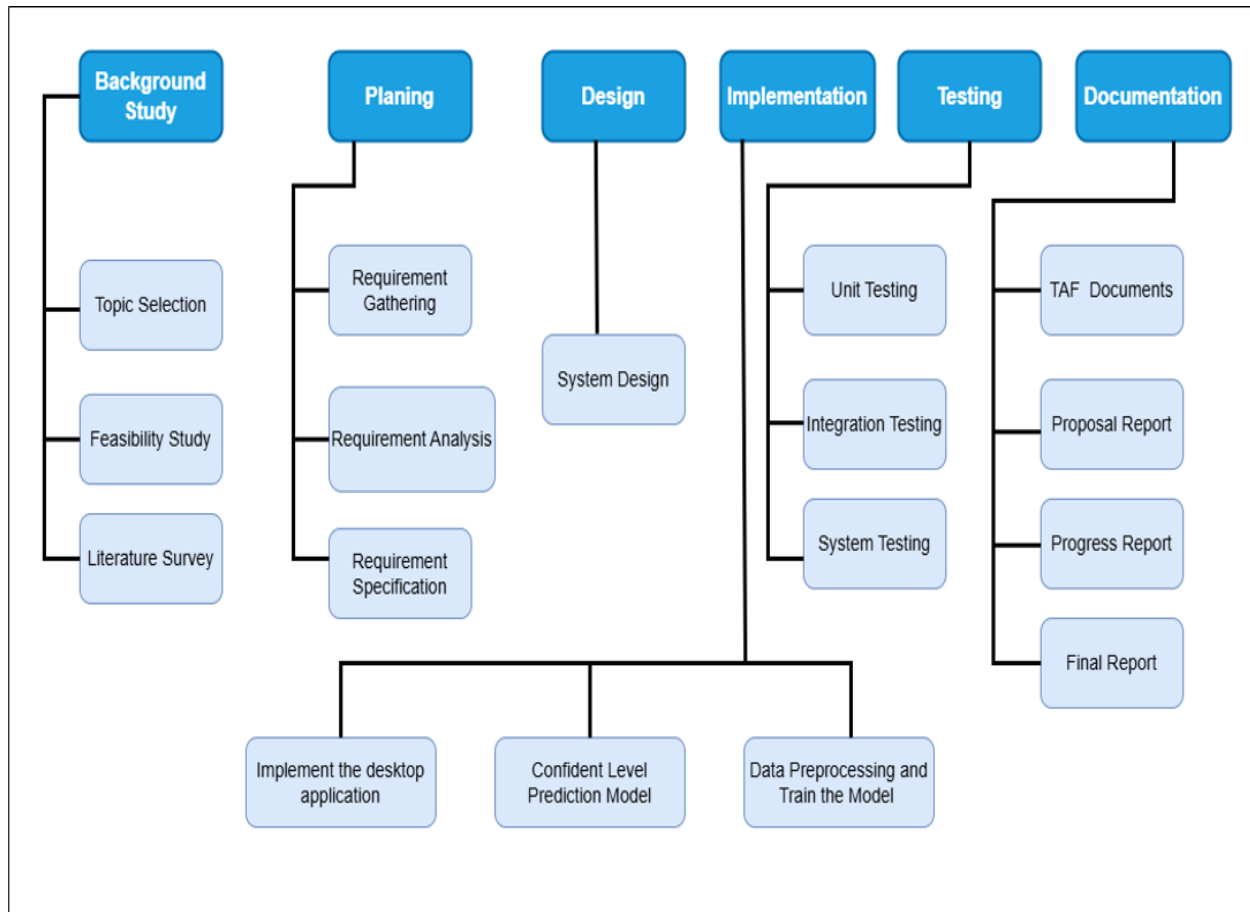


Figure 5. Work Break Down Structure

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- Incident pattern analysis.
- Clustering and classifications.
- Semantic testimony analysis.
- Alert system.
- Realtime updates.

4.2 Non- Functional Requirements

- Privacy and Security.
- Reliability.
- Usability.
- Compatibility.
- Accuracy.

4.3 System Requirements

- **Web-Based Interface:** A user-friendly React-based interface that allows investigators to interact with crime data, visualize case relationships, and access pattern analysis insights.
- **Machine Learning Models:** Utilizes clustering, classification, and NLP algorithms to detect patterns and correlations between incidents.
- **Incident Correlation & Real-Time Alerts:** Automatically links similar cases and sends **notifications** on emerging crime patterns.
- **Data Storage & Retrieval:** Secure PostgreSQL/MongoDB database for storing case data, testimonies, and investigative reports.
- **Cloud Services for Scalability:** Cloud-based infrastructure (AWS, Google Cloud) for efficient data processing and model execution.

4.4 User Requirements

- **Law Enforcement Experts:** Collaboration with investigators and forensic analysts to refine crime correlation models.
- **Criminal Justice Institutions:** Possible integration with police departments, crime labs, and legal institutions to improve investigative workflows.

5. COMMERCIALIZATION

5.1 Market Opportunity

Market Gap

- Existing interrogation tools lack AI-driven dynamic questioning, real-time behavioral analysis, and multi-modal data processing (audio, video, text).
- Current systems focus on static analysis (e.g., lie detection, speech transcription) but fail to generate adaptive, context-aware follow-up questions for investigators.
- Manual interrogation methods lead to investigator bias, fatigue-related errors, and missed insights, creating inefficiencies in legal and forensic investigations.

Target Users

- Law enforcement agencies – Police departments, national security agencies, forensic teams.
- Private investigators – Individual and agency-level investigators handling fraud, corporate investigations, and legal cases.
- Judicial bodies – Courts, public defenders, prosecution teams using AI-assisted testimony evaluation.

5.2 Value Proposition

- Automates data handling and incident correlation, significantly reducing human error and time spent on manual interrogation analysis.
- Provides real-time behavioral and emotional insights, helping investigators detect stress, deception, and inconsistencies in suspect testimonies.
- Generates dynamic, AI-driven follow-up questions based on suspect responses, contradictions, and detected emotional cues.
- Reduces investigator workload by providing automated response analysis, contradiction detection, and question optimization.
- Predictive analytics for case resolution – Helps law enforcement identify patterns, cross-reference historical case data, and improve investigative efficiency.

5.3 Revenue Model

Subscription-Based Model (SaaS - Monthly/Yearly)

- Suitable for law enforcement agencies and investigative firms requiring continuous access.
- Pricing tiers based on number of users, AI processing capacity, and integrated behavioral analysis features.

Per-Case Usage Pricing

- Designed for independent investigators, legal consultants, and forensic analysts who require on-demand AI interrogation assistance.
- Charges based on number of interrogations, AI-generated questions, and behavioral analysis reports per case.

Enterprise Licensing for Large-Scale Deployment

- For national security agencies, forensic research institutions, and high-profile investigation units.
- Custom contracts for on-premise deployment, advanced AI customization, and dedicated support.

Additional Monetization Opportunities

- API Integration – Licensing AI interrogation capabilities for third-party forensic and legal tech platforms.
- Partnerships with law enforcement bodies for custom AI model training and integration.
- Expansion into multilingual markets to offer global AI-driven investigative tools.

6. REFERENCES

1. A. T. A. Sadiqur Rahman and A. Kaur, "The Application of Artificial Intelligence in the Administration of Criminal Law and the Procedures Involved in Criminal Cases," *2024 International Conference on Advances in Computing Research on Science Engineering and Technology (ACROSET)*, Indore, India, 2024, pp. 1-5. doi: 10.1109/ACROSET62108.2024.10743336.
2. V. Singh, A. Dixit, S. Pandey, B. V. Kumar, V. Pachouri, and M. Sahu, "Role of Artificial Intelligence in Criminal Investigation," *2023 International Conference on Quantum Technologies, Communications, Computing, Hardware and Embedded Systems Security (iQ-CCHES)*, Kottayam, India, 2023, pp. 1-4. doi: 10.1109/iQ-CCHES56596.2023.10391288.
3. J. Baltrūnienė, "Place of artificial intelligence in the detection and investigation of crime: the present state and future perspectives," *Current Problems of Forensic Science*, vol. 26, pp. 43-58, 2023. doi: 10.52097/pwk.5431.
4. I. Lakshan, L. Wickramasinghe, S. Disala, S. Chandrasegar, and P. S. Haddela, "Real-Time Deception Detection for Criminal Investigation," *2019 National Information Technology Conference (NITC)*, Colombo, Sri Lanka, 2019, pp. 90-96. doi: 10.1109/NITC48475.2019.9114422.
5. R. Ejjami, "AI-driven Justice: Evaluating the Impact of Artificial Intelligence on Legal Systems," *IJFMR*, vol. 6, no. 3, pp. 1-6, May-June 2024. doi: 10.36948/ijfmr.2024.v06i03.23969.
6. S. S. Das, S. Patnaik, P. Pattnayak, and A. Mohanty, "An Advancement in Forensic and Criminal Investigation Through Artificial Intelligence," *2023 OITS International Conference on Information Technology (OCIT)*, Raipur, India, 2023, pp. 920-926. doi: 10.1109/OCIT59427.2023.10431257.
7. J. K. Sabherwal and K. Kaur, "The Role of Artificial Intelligence in Improving Criminal Justice System," *International Journal of Science and Research (IJSR)*, vol. 13, no. 4, pp. 999-1002, April 2024. doi: 10.21275/SR24412153835.
8. Y. Hu, Y. Li, T. Yang, and Q. Pan, "Short Text Classification with A Convolutional Neural Networks Based Method," *2018 15th International Conference on Control, Automation, Robotics and Vision (ICARCV)*, Singapore, 2018, pp. 1432-1435. doi: 10.1109/ICARCV.2018.8581332.
9. S. Lai, L. Xu, K. Liu, and J. Zhao, "Recurrent Convolutional Neural Networks for Text Classification," *AAAI*, vol. 29, no. 1, Feb. 2015.
10. A. Conneau, H. Schwenk, L. Barrault, and Y. Lecun, "Very Deep Convolutional Networks for Text Classification," in *Proceedings of the 15th Conference of the*

European Chapter of the Association for Computational Linguistics: Volume 1, Long Papers, Valencia, Spain, 2017, pp. 1107-1116.

11. A. Higginson, E. Eggins, and L. Mazerolle, "Police Techniques for Investigating Serious Violent Crime: A Systematic Review," *Trends and Issues in Crime and Criminal Justice*, vol. 539, 2017.

12. C. Rigano, "Using Artificial Intelligence to Address Criminal Justice Needs," *National Institute of Justice (NIJ)*, Oct. 8, 2018. [Online]. Available: <https://nij.ojp.gov/topics/articles/using-artificial-intelligence-address-criminal-justice-needs>. [Accessed: Dec. 4, 2024].

13. "Investigation and Evidence Gathering," *Criminal Justice Research*, [Online]. Available: <https://criminal-justice.iresearchnet.com/criminal-justice-process/investigation-and-evidence-gathering>. [Accessed: Dec. 4, 2024].

14. EclipseForensics, "Comparing Traditional vs. Modern Forensic Methods: A Brief Guide," *EclipseForensics*, Oct. 2, 2024. [Online]. Available: <https://eclipseforensics.com/comparing-traditional-vs-modern-forensic-methods-a-brief-guide>. [Accessed: Dec. 4, 2024].

15. "Sri Lanka Law, Code of Criminal Procedure Act," [Online]. Available: <https://www.srilankalaw.lk/c/217-code-of-criminal-procedure-act.html>. [Accessed: Dec. 5, 2024].

Appendix

Turnitin Report

